BUSINESS INTELLIGENCE LAB

EXPERIMENT 6

Dashboard and Scorecard Development using Microsoft Power BI

Problem file

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1. Aim

To design and develop an interactive Executive Dashboard and Business Scorecard using Microsoft Power BI for monitoring organizational performance, supporting business decision-making, and communicating key performance indicators (KPIs) effectively.

2. Tools / Software Required

1. Microsoft Power BI Desktop
2. Global Superstore Dataset.xlsx
3. Power Query Editor
4. DAX (Data Analysis Expressions)

3. Procedure

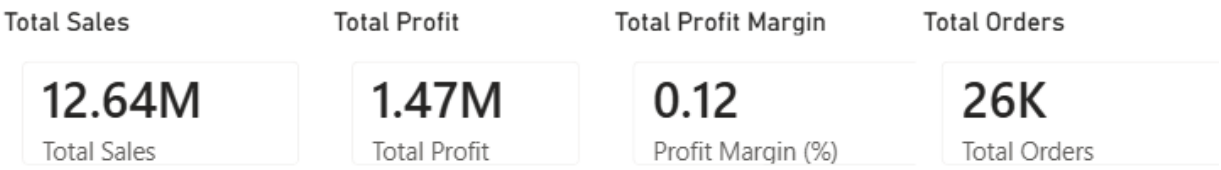
Part A: Dashboard Planning

Step 1: Open Microsoft Power BI Desktop and open the existing Power BI project created in the previous experiments.

Step 2: Identify the important business metrics required by the management team, such as Total Sales, Total Profit, Total Orders, and Profit Margin.

Step 3: Select suitable visualizations for representing different business performance indicators.

Step 4: Plan the dashboard layout by deciding the placement of KPI cards, charts, slicers, and other visual elements.



Part B: Create KPI Dashboard

Step 5: Create KPI Cards for the following metrics:

- Total Sales
- Total Profit
- Total Orders
- Profit Margin (%)

Step 6: Create the following visualizations:

1. Sales by Region
2. Sales by Category
3. Monthly Sales Trend
4. Top 10 Customers by Sales
5. Profit by Segment

Step 7: Format all the visualizations by applying suitable colors, titles, labels, fonts, and alignment to improve readability and maintain consistency throughout the dashboard.



Part C: Add Interactive Features

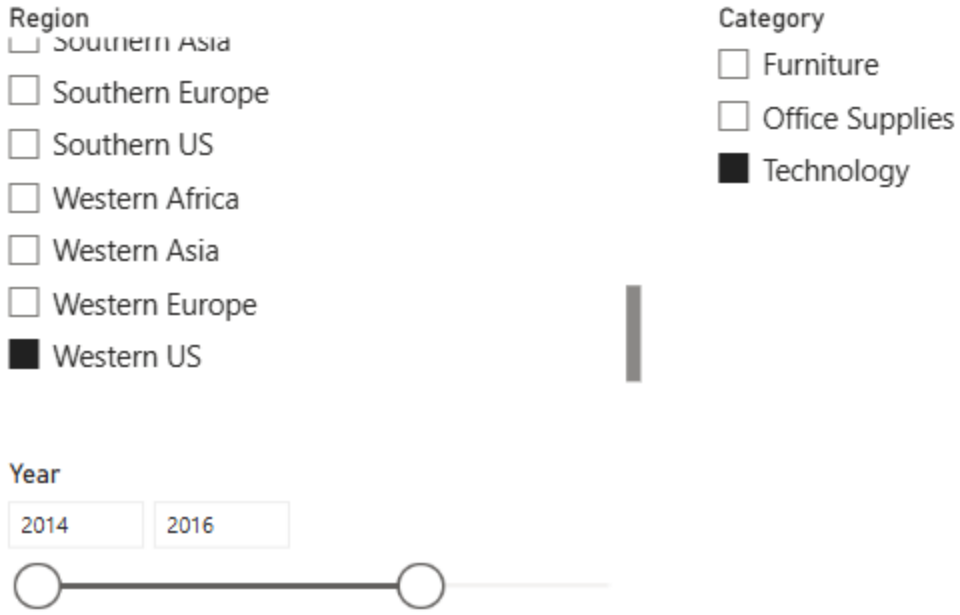
Step 8: Add slicers to the dashboard for the following fields:

- Region
- Category
- Year

Step 9: Configure the slicers so that the dashboard visuals respond dynamically whenever a user selects a particular region, category, or year.

Step 10: Test the dashboard interactivity by selecting different slicer values and verifying that the KPI cards and charts update accordingly.

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Part D: Create Business Scorecard

Step 11: Create a Business Scorecard containing the following performance comparisons:

- Sales Target vs Actual Sales
- Profit Target vs Actual Profit
- Order Target vs Actual Orders

Step 12: Apply conditional formatting to highlight the performance status of each KPI.

Step 13: Use suitable performance indicators to represent the status of each target:

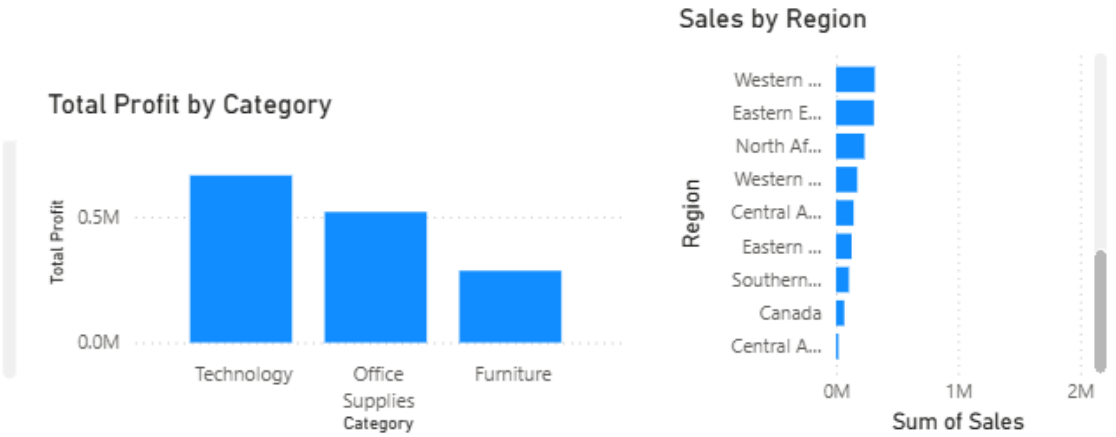
- Achieved
- Partially Achieved
- Not Achieved

Part E: Business Analysis

Step 14: Analyze the dashboard findings by examining the KPI cards and different visualizations.

Step 15: Identify the following business performance indicators:

1. Best Performing Region : Western Europe
2. Worst Performing Region : Central Asia
3. Most Profitable Category : Technology
4. Least Profitable Category : Furniture



Part F: Final Verification

Step 17: Verify that all KPI cards, charts, slicers, and scorecard elements are properly displayed.

Step 18: Check that all interactive features work correctly and that the dashboard provides accurate and meaningful information.

Step 19: Save the completed Power BI project for future reference and analysis.



4. Suggested DAX Measures

The following DAX measures can be used to calculate the required KPIs. Adjust table and column names according to your Power BI dataset.

1. Total Sales

Total Sales = SUM(Orders[Sales])

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2. Total Profit

Total Profit = SUM(Orders[Profit])

3. Total Orders

Total Orders = DISTINCTCOUNT(Orders[Order ID])

4. Profit Margin (%)

Profit Margin (%) =
DIVIDE([Total Profit], [Total Sales], 0)

Format the Profit Margin measure as Percentage.

6. Result

The interactive Executive Dashboard and Business Scorecard were successfully developed using Microsoft Power BI. The dashboard displays important business KPIs through different visualizations and provides interactive filtering options using slicers. The Business Scorecard enables comparison of actual performance against predefined targets, helping management monitor organizational performance and make data-driven decisions.